

Ketazine-Linked Crystalline Porous Covalent Organic Frameworks

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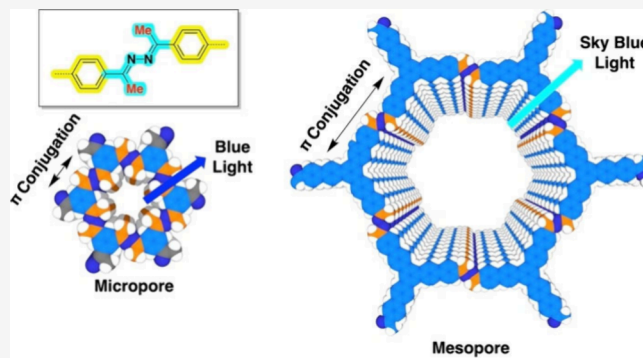


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Supporting Information

ABSTRACT: Covalent organic frameworks (COFs) are a class of crystalline porous materials with well-defined π arrays and ordered channels, which can be predesigned with a topology diagram and prepared via a polycondensation reaction. Over the past decade, various types of π building units with different functional groups have been developed. Among them, aldehyde is one of the most widely used monomers that form COFs with azine, hydrazine, imine, squaraine, and C=C linkages. In contrast, its closest analogue, i.e., ketone, despite its broad structural diversity, has not yet been investigated for the design and synthesis of COFs. Herein we report the first examples of ketazine-linked COFs by developing ketones as monomers to enable polycondensation with hydrazine under solvothermal conditions. We observed that a careful screening of reaction conditions including solvent, catalyst, concentration, reaction temperature, and reaction time leads to the finding of optimal polymerization systems to produce highly crystalline and porous ketazine-linked COFs. Surprisingly, the ketazine linkage enables π conjugation between knot and linker sites and renders the resultant materials able to emit a strong blue fluorescence, highlighting the π electronic features of this new family of COFs. Our findings of ketones as monomers and ketazine as linkage bring unprecedented structures, functions, and applications to the field of COFs.



INTRODUCTION

Covalent organic frameworks (COFs) are a class of crystalline porous materials, which develop extended structures along both in-plane and out-of-plane directions.^{1–4} Progress in chemistry over the past decade has enabled the design and development of various π systems as building blocks for the construction of diverse COFs, greatly expanding and deepening our understanding of this class of emerging materials and the capability of designing and controlling their structures and properties. Especially the linkage chemistry, i.e., the monomer backbone structure and the type of reactive groups, is crucial for the polymerization reaction, the structural formation, and the development of property and function.⁵ Among various monomers, aldehyde is one of the most widely used monomers which generates azine,^{6–10} imine,^{11–13} hydrazone,^{14–16} squaraine,^{17,18} and C=C linked COFs.^{19–21} However, its closest analogue, i.e., monoketone, although bearing a broad structural diversity, has not yet been developed for the construction of COFs.

Herein, we report the first examples of ketazine-linked COFs by developing ketones as monomers for polycondensation with hydrazine under solvothermal conditions. We designed 1,1',1''-(benzene-1,3,5-triyl)triethanone (TKB) and 1,1',1''-(benzene-1,3,5-triyltris([1,1'-biphenyl]-4',4'-diyl))tris(ethan-1-one) (TKBPB) as ketone monomers for the condensation with hydrazine to produce ketazine-linked crystalline microporous and mesoporous COFs (Schemes 1A and 1B).

RESULTS AND DISCUSSION

Design. To showcase the strategy, we focused on the most general topology—hexagonal topology—by designing C_3 -symmetric knots with different sizes to enable the construction of both microporous and mesoporous COFs (Schemes 1A and 1B).

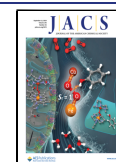
Synthesis of Mesoporous TKBPB-Ketazine-COF. To synthesize ketazine-linked COFs, we systematically investigated the polycondensation reaction between TKBPB and hydrazine by tuning solvent, catalyst, reaction temperature, and time. We first tried conventional solvothermal conditions using acetic acid (6 M, 100 μ L) as the catalyst in various solvents, such as *o*-dichlorobenzene (*o*-DCB), dioxane, *m*-xylene, benzonitrile, anisole, *o*-DCB/*n*-butanol (*n*-BuOH), *m*-xylene/*n*-BuOH, benzonitrile/*n*-BuOH, anisole/*n*-BuOH, and mesitylene/dioxane. However, polymerization of TKBPB with hydrazine at 120 °C yields only solutions even after a 5 day reaction, where no reaction occurs under these conditions (Table S1).

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Scheme 1. (A, B) Schematic of the Synthesis of Ketazine-Linked (A) Microporous TKB-Ketazine-COF and (B) Mesoporous TKBPB-Ketazine-COF; (C, D) Reconstructed Crystal Structure of (C) Single Layer and (D) Ten Layers of TKB-Ketazine-COF at Top View. (E, F) Reconstructed Crystal Structure of (E) Single Layer and (F) Ten Layers of TKBPB-Ketazine-COF at Top View (in Parts C–F, C = Blue, N = Orange, and White = H)

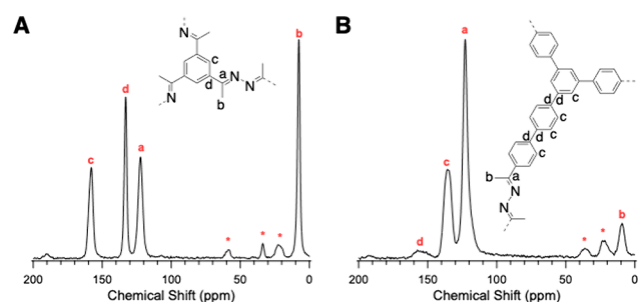
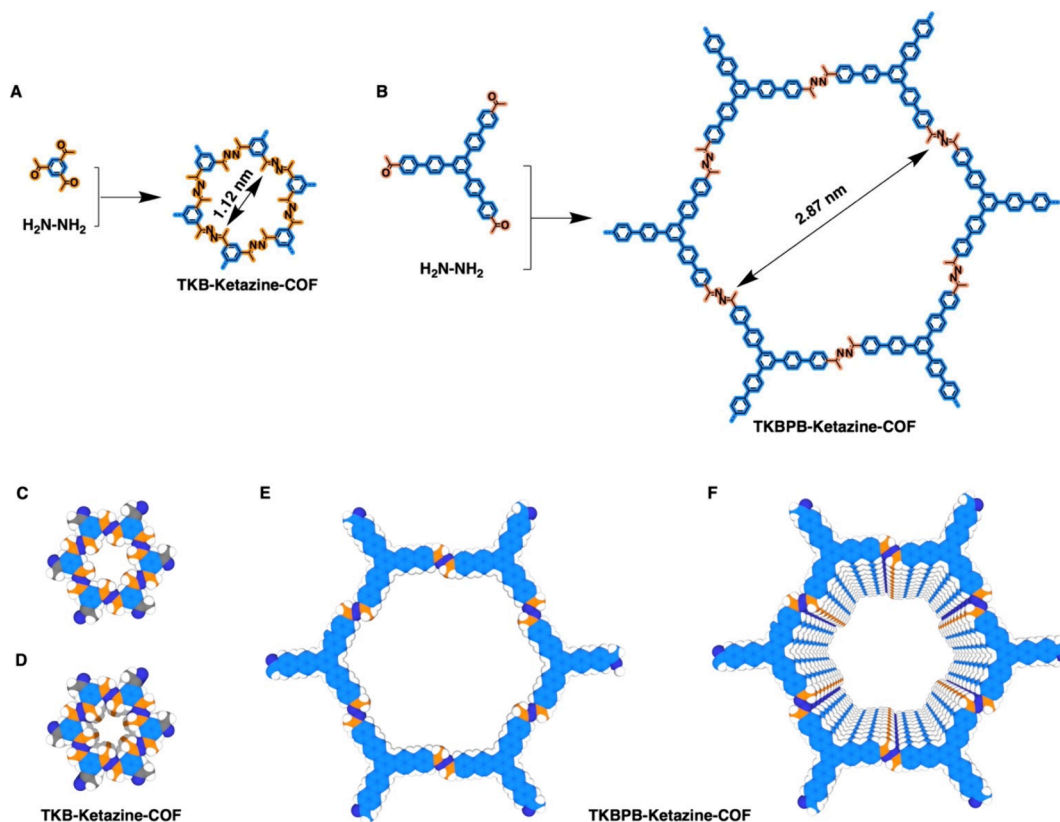


Figure 1. ^{13}C CP/MAS spectra of (A) TKB-Ketazine-COF and (B) TKBPB-Ketazine-COF.

Next, we screened the concentration of acetic acid from 6 M to pure acetic acid in different solvents (Table S2). Among the different concentrations of acetic acid, only the systems with pure acetic acid yielded solids, while other acetic acid concentrations resulted in solutions (Table S2). We observed that solvents such as *o*-DCB/*n*-BuOH (1/1, vol.), mesitylene/dioxane (1/1, vol.), and anisole/*n*-BuOH (1/1, vol.) in the presence of pure acetic acid generated TKBPB-Ketazine-COF with medium intensities of powder X-ray diffraction (PXRD) (Figure S1) and achieved yields of 68%, 72%, and 80%, respectively, while using dioxane as a solvent results in a weak PXRD intensity and a low yield of 62%. These results indicate that the anisole/*n*-BuOH (1/1, vol.) system in the presence of pure acetic acid is preferred but also suggest that polymerization proceeds to a certain degree but stops at an early stage of low conversions. We further tuned the amount of pure

acetic acid from 30 to 80 μL and observed that the anisole/*n*-BuOH (1/1, vol.) system with 50 or 60 μL of acetic acid affords the highest PXRD intensity after a 5 day reaction (Table S3 and Figure S2). As the anisole/*n*-BuOH system leads to the generation of TKBPB-Ketazine-COF, we tuned the solvent ratio from 6/1 to 1/6 (Table S4 and Figure S3). We found that the anisole/*n*-BuOH (4/1, vol.) system leads to the highest PXRD intensity and 82% yield. Moreover, screening reaction temperatures at 90, 120, and 150 $^{\circ}\text{C}$ revealed that 120 $^{\circ}\text{C}$ affords the highest PXRD intensity (Table S5 and Figure S4). Extension of reaction time from 1 to 7 days disclosed that after 5 days the PXRD intensity became unchanged (Table S6 and Figure S5). On the other hand, using pure trifluoroacetic acid (TFA, 30–70 μL) or TFA solution (0.5 M, 70 or 90 μL) as a catalyst also yields crystalline TKBPB-Ketazine-COF but decreases the PXRD intensity (Table S7 and Figure S6). Notably, the PXRD pattern of TKBPB-Ketazine-COF is different from that of the TKBPB monomer (Figure S7). With these results, the optimal polymerization conditions are found to be TKBPB (19.8 mg, 30 μmol), hydrazine (2.2 μL , 45 μmol), anisole/*n*-BuOH (4/1, 2 mL), and pure acetic acid (60 μL), 120 $^{\circ}\text{C}$, and 5 days. We emphasize that the polymerization under this condition is reproducible to produce highly crystalline TKBPB-Ketazine-COF in 80%–85% yields.

Synthesis of Microporous TKB-Ketazine-COF. With the successful synthesis of large-pore TKBPB-Ketazine-COF, we applied its optimal conditions to the synthesis of small-pore TKB-Ketazine-COF. However, these optimal conditions

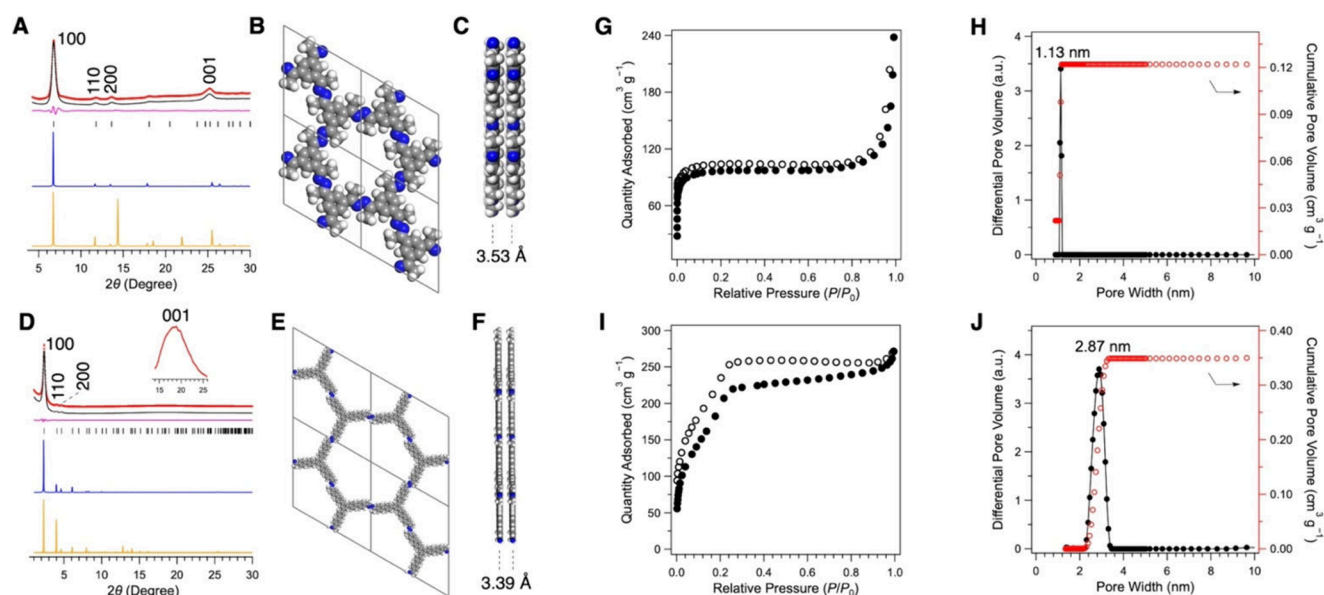


Figure 2. (A) PXRD of experimentally observed (red cross), Pawley-refined (black curve), and their difference (pink curve), simulated from the eclipsed AA-stacking mode (blue curve), AB-stacking mode (orange curve), and Bragg position (black bar) of TKB-Ketazine-COF. (B, C) Structure of (B) unit cell and (C) interlayer distance of TKB-Ketazine-COF. (D) PXRD of experimentally observed (red cross), Pawley-refined (black curve), and their difference (pink curve), simulated from the eclipsed AA-stacking mode (blue curve), AB-stacking mode (orange curve), and Bragg position (black bar) of TKBPB-Ketazine-COF. (E, F) Structure of (E) unit cell and (F) interlayer distance of TKBPB-Ketazine-COF. (G, H) N_2 sorption isotherm (G) and pore size (H, black dots) and distribution (red dots) of TKB-Ketazine-COF. (I, J) N_2 sorption isotherm (I) and pore size (J, black dots) and distribution (red dots) of TKBPB-Ketazine-COF.

resulted in only clear solutions, irrespective of acetic acid concentration and solvent combination (Table S8). Thus, a systematic condition screening using acetic acid as the catalyst was performed (Table S8). Unfortunately, these conditions gave rise to solutions or gels, irrespective of acetic acid concentration, mixture of solvents, reaction temperature, and reaction time, where no reaction or limited reaction occurs under these conditions (Table S8). Drying gels by vacuum results in solids, which, however, did not show any PXRD patterns (Figure S8). Thus, we screened the type of acids as catalysts in the anisole/*n*-BuOH (3/1, vol.) system. Among acetic acid, TFA, and *p*-toluenesulfonic acid (PTSA), TFA leads to the highest PXRD intensity, while PTSA results in solutions (Table S9 and Figure S9). We tuned the concentration of TFA and observed that the concentration at 0.5 M results in clear solutions, while the system with pure TFA as catalyst affords TKB-Ketazine-COF with high crystallinity (Table S9 and Figure S9).

To further improve the PXRD intensity, we employed different amounts of monosubstituted acetophenone as a modulator with an acetophenone/TKB molar ratio of 1–6 for the polymerization (Table S10 and Figure S10). As a result, acetophenone at a molar ratio of 3 to TKB greatly increased the PXRD intensity. Based on these results, we screened the solvent, reaction temperature, and reaction time (Table S11 and Figures S11–S14). Consequently, the optimal condition for the synthesis of TKB-Ketazine-COF was found to be TKB (20 mg, 98 μ mol), hydrazine (7.1 μ L, 45 μ mol), and acetophenone (34.3 μ L, 294 μ mol) in anisole/*n*-BuOH (3/1 vol., 2 mL) with pure TFA (60 μ L) at 120 $^{\circ}$ C for 5 days. This condition can reproducibly form TKB-Ketazine-COF in 71%–86% yield (Figure S13). Notably, the PXRD pattern of TKB-Ketazine-COF is different from that of the TKB monomer (Figure S15).

Characterization. Fourier transform infrared (FT-IR) and solid-state ^{13}C cross-polarization magic angle spinning nuclear magnetic resonance (CP/MAS NMR) spectroscopies confirmed the chemical structures of ketazine-linked COFs. FT-IR displayed a vibrational band at 1630 cm^{-1} that was assigned to the $\text{C}=\text{N}$ bonds of TKB-Ketazine-COF (Figure S16, red curve) and TKBPB-Ketazine-COF (Figure S17, red curve), while the band at 1361 cm^{-1} was assigned to the $\text{C}-\text{H}$ (CH_3) bond. This was verified by their model compounds (Figures S16 and S17, blue curves). In the NMR spectra, typical signals at 122.9 and 122.4 ppm were assigned to the carbons of the $\text{C}=\text{N}$ linkages, and the signals at 9.31 and 7.67 ppm were allocated to the methyl groups of TKB-Ketazine-COF (Figure 1A) and TKBPB-Ketazine-COF (Figure 1B), respectively. Elemental analyses revealed that the C, H, and N contents agree with their theoretical values (Table S12).

The crystalline structures of TKB-Ketazine-COF and TKBPB-Ketazine-COF were determined by PXRD analysis and computational simulations. TKB-Ketazine-COF (Figure 2A, red cross) exhibited the strongest PXRD peak at 6.77 $^{\circ}$, which was assigned to the reflection from the (100) facet. The other PXRD peaks at 11.74 $^{\circ}$ and 13.63 $^{\circ}$ were assigned to the (110) and (200) facets, respectively. The presence of the PXRD peak at 25.15 $^{\circ}$ assigned to the (001) facet indicates that TKB-Ketazine-COF is ordered along the *z* direction with an interlayer interval of 3.539 $\text{\AA}$. The Pawley refinement (Figure 2A, black curve) reproduced the experimental data with a negligible difference (Figure 2A, pink curve), $R_{\text{WP}} = 5.01\%$, and $R_p = 3.33\%$. We have simulated the system using the *P6/m* space group first (Table S13) and observed that the system finally converged to yield a space group of *P1*. The Rietveld refinement showed that a space group of *P1* with lattice parameters of $a = 14.9828 \text{ \AA}$, $b = 15.0122 \text{ \AA}$, $c = 3.7660 \text{ \AA}$, $\alpha = 90.5199^{\circ}$, $\beta = 90.5104^{\circ}$, and $\gamma = 120.083^{\circ}$ can reproduce the PXRD pattern (Figure 2A, blue curve), while the AB-stacking

mode (Figure 2A, orange curve) does not reproduce the experimentally observed result. Thus, TKB-Ketazine-COF assumes a crystalline hexagonal structure with an AA-stacking mode to form extended two-dimensional (2D) polymers (Figure 2B) and layered frameworks (Figure 2C).

TKBPB-Ketazine-COF displayed a strong PXRD peak at 2.36° , which was assigned to the reflection from the (100) facet (Figure 2D, red cross). The peaks at 3.99° and 4.65° were assigned to the (110) and (200) facets, respectively. The PXRD peak at 18.25° assigned to the (001) facet suggests that TKBPB-Ketazine-COF possesses an interlayer interval of 4.858 Å along the *z* direction. We have simulated the system using the *P6/M* space group first (Table S14), while the system converged to show a space group of *P1*. The Pawley refinement (Figure 2D, black curve) regenerated the experimental profile with a small difference (Figure 2D, pink curve), with $R_{\text{WP}} = 2.33\%$ and $R_p = 1.64\%$. Rietveld refinement revealed that a space group of *P1* with lattice parameters of $a = 44.4519$ Å, $b = 44.3809$ Å, $c = 3.8004$ Å, $\alpha = 88.1460^\circ$, $\beta = 92.4973^\circ$, and $\gamma = 119.430^\circ$ reproduces the PXRD pattern (Figure 2D, blue curve), while the AB-stacking mode (Figure 2D, orange curve) hardly reproduces the observed PXRD result. Consequently, TKBPB-Ketazine-COF adopts a crystalline hexagonal structure to develop extended 2D layers (Figure 2E) and AA-stacking frameworks (Figure 2F).

Porosity. Nitrogen sorption experiments of TKB-Ketazine-COF at 77 K revealed a type I isotherm with a sharp uptake at a low pressure of $P/P_0 < 0.05$, which is characteristic of microporous materials (Figure 2G). The Brunauer–Emmett–Teller (BET) surface area was evaluated to be $340 \text{ m}^2 \text{ g}^{-1}$, with a pore volume of $0.121 \text{ cm}^3 \text{ g}^{-1}$ (Figure 2H). The pore size is 1.13 nm (Figure 2H and Table S15), which agrees with the theoretical value. In contrast, TKBPB-Ketazine-COF exhibited a type IV isotherm, which is characteristic of mesoporous materials (Figure 2I). The BET surface area was evaluated as $702 \text{ m}^2 \text{ g}^{-1}$, with a pore volume of $0.35 \text{ cm}^3 \text{ g}^{-1}$ (Figure 2J). The pore size was 2.87 nm (Figure 2J), which is consistent with that of the lattice.

We analyzed the morphology by dispersing the ketazine-linked COFs in ethanol and carried out field emission scanning electron microscopy (FE SEM). TKB-Ketazine-COF (Figure S18) adopts a rod-like morphology, and TKBPB-Ketazine-COF (Figure S19) appears as plate-like objects. High-resolution transmission electron microscopy (HR TEM) revealed an ordered structure of TKB-Ketazine-COF (Figure S20).

Stability. Noticeably, ketazine-linked COFs are both thermally and chemically robust. Thermogravimetric analysis (TGA) revealed that TKB-Ketazine-COF (Figure S21) and TKBPB-Ketazine-COF (Figure S22) are thermally stable up to 350°C under nitrogen. To test their chemical stability, TKB-Ketazine-COF and TKBPB-Ketazine-COF were immersed in different solvents, including tetrahydrofuran (THF), dimethylformamide (DMF), methanol, boiling water, aqueous HCl (6 M), and aqueous NaOH (6 M). After one-week treatment, the samples were rinsed, dried, and subjected to FT-IR and PXRD measurements (Figure 3A–E). The retained C=N bond in the FT-IR spectra, unaltered position and intensity of the PXRD peaks, and less weight loss (Figure 3E,F) demonstrated that their skeletons and crystalline structures are retained under these conditions.

Light Adsorption and Luminescence in Solutions. The ketazine linkage $-(\text{CH}_3)\text{C}=\text{N}-\text{N}=\text{C}(\text{CH}_3)-$ con-

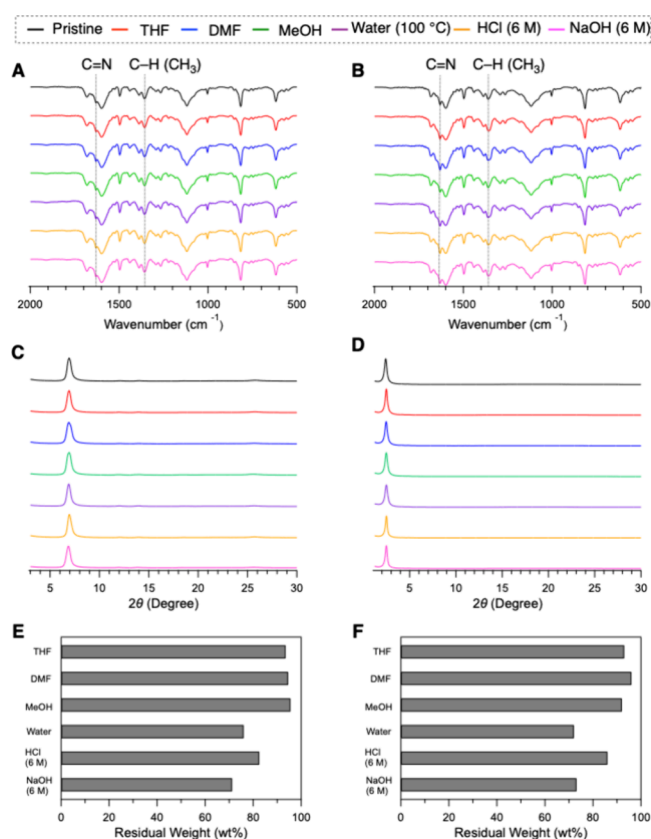


Figure 3. Stability: (A, B) FT-IR spectra of (A) TKB-Ketazine-COF and (B) TKBPB-Ketazine-COF after treated under different conditions. (C, D) PXRD patterns of (C) TKB-Ketazine-COF and (D) TKBPB-Ketazine-COF after treated under different conditions. (E, F) Weight loss of (E) TKB-Ketazine-COF and (F) TKBPB-Ketazine-COF after treated under different conditions.

sists of two C=N bonds mediated by one N–N bond. To elucidate if the ketazine linkage allows π electronic conjugation, we deployed electronic absorption spectroscopy to investigate their absorption band. For this purpose, we dispersed TKB-Ketazine-COF and TKBPB-Ketazine-COF in THF and carried out exfoliation with the super sonication technique. Dynamic light scattering (DLS) analysis revealed that TKB-Ketazine-COF and TKBPB-Ketazine-COF exhibited one DLS signal with a particle size of 200–300 nm (Figures S23 and S24). TKB-Ketazine-COF in THF displayed an absorption band at 324 nm (Figure 4A, solid blue curve; Table S16), which is 32 nm red-shifted from that of the monomer TKB (Figure 4A dotted black curve, 292 nm). On the other hand, TKBPB-Ketazine-COF in THF exhibited its absorption band at 415 nm (Figure 4B, solid red curve), which is 107 nm red-shifted from that of TKBPB (Figure 4B, dotted black curve, 308 nm). These red-shifted absorption bands suggest that the ketazine linkage enables a higher degree of π delocalization in the extended 2D frameworks.

The capability of extending π conjugation by the ketazine linkage triggers interest in exploring their light-emitting behaviors. TKB-Ketazine-COF in THF, upon excitation at 324 nm, emits a brilliant blue light centered at 450 nm (Figure 4C, solid blue curve). In contrast, the monomer TKB emits a fluorescence at 437 nm (Figure 4C, dotted blue curve). Notably, TKB-Ketazine-COF displays an absolute fluorescence quantum yield as high as $15.5 \pm 1.5\%$ (Figure 4D and Table

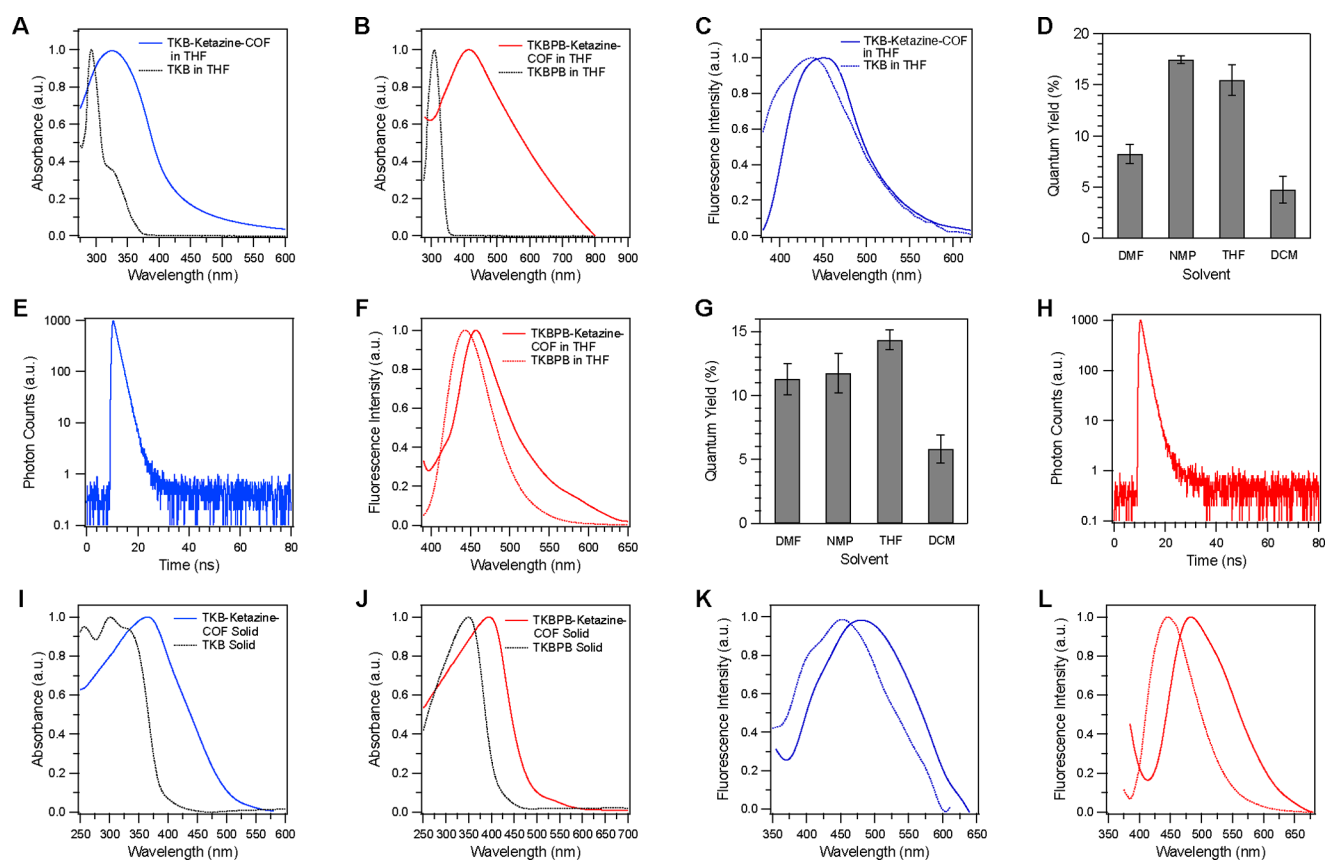


Figure 4. (A) Absorption spectra of TKB (dotted black curve) and TKB-Ketazine-COF (solid blue curve) in THF. (B) Absorption spectra of TKBPB (dotted black curve) and TKBPB-Ketazine-COF (solid red curve) in THF. (C) Fluorescence spectra of TKB (dotted blue curve) and TKB-Ketazine-COF (solid blue curve) in THF. (D) Absolute fluorescence quantum yield of TKB-Ketazine-COF in different solvents. (E) Transient decay profile of TKB-Ketazine-COF in THF. (F) Fluorescence spectra of TKBPB (dotted red curve) and TKBPB-Ketazine-COF (solid red curve) in THF. (G) Absolute fluorescence quantum yield of TKBPB-Ketazine-COF in different solvents. (H) Transient decay profile of TKBPB-Ketazine-COF in THF. (I) Solid-state absorption spectra of TKB (dotted black curve) and TKB-Ketazine-COF (solid blue curve). (J) Solid-state absorption spectra of TKBPB (dotted black curve) and TKBPB-Ketazine-COF (solid red curve). (K) Solid-state fluorescence spectra of TKB (dotted blue curve) and TKB-Ketazine-COF (solid blue curve). (L) Solid-state fluorescence spectra of TKBPB (dotted red curve) and TKBPB-Ketazine-COF (solid red curve).

S16). We measured the fluorescence lifetime by a time-resolved photon counting method and observed that TKB-Ketazine-COF possesses a fluorescence lifetime of 1.74 ns (Figure 4E). TKBPB-Ketazine-COF emits strong blue luminescence at 457 nm (Figure 4F, solid red curve), with an absolute fluorescence quantum yield of $14.4 \pm 0.8\%$ (Figure 4G) and a fluorescence lifetime of 1.67 ns (Figure 4H and Table S16). In contrast, the TKBPB monomer emitted a fluorescence band at 444 nm (Figure 4F, dotted red curve). These interesting and novel results originate from the 2D ketazine-linked, yet π -conjugated frameworks.

When dispersed in other solvents such as dichloromethane (DCM), dimethylformamide (DMF), and *N*-methylpyrrolidone (NMP), TKB-Ketazine-COF emits brilliant blue light at 458, 464, and 465 nm, with a quantum yield of $4.8 \pm 1.3\%$, $8.3 \pm 0.9\%$, and $17.5 \pm 0.4\%$, respectively (Figure 4D, Figures S25–S30, and Table S16). On the other hand, TKBPB-Ketazine-COF emits blue at 447, 452, and 448 nm with a quantum yield of $5.8 \pm 1.1\%$, $11.3 \pm 1.2\%$, and $11.7 \pm 1.6\%$ in DCM, DMF, and NMP, respectively (Figure 4G, Figures S31–S36, and Table S16). Fluorescence decay files revealed that their lifetimes in different solvents are between 1.82 and 9.24 ns (Figures S37–S42 and Table S16). Notably, TKB-Ketazine-COF exhibited an extremely long lifetime of

7.75 ns in NMP, while TKBPB-Ketazine-COF displayed a long lifetime of 9.24 ns in DMF. These exceptional lifetimes are rare and abnormal in COFs and reflect that ketazine-linked COFs retain a photoexcited state over a long period. The comparison of blue-light-emitting COFs was summarized in Table S17.

Solid-State Light Absorption and Luminescence.

Along this line, we further investigated the solid-state light absorption and light-emitting properties. The solid sample of TKB displays an absorption band at 329 nm (Figure 4I, dotted black curve), while TKB-Ketazine-COF exhibits a 35 nm red-shifted absorption band at 364 nm (solid blue curve). On the other hand, the TKBPB solid sample shows an absorption band at 349 nm (Figure 4J, dotted black curve), which is 45 nm red-shifted to 394 nm for TKBPB-Ketazine-COF (solid red curve). The large red-shifts by 35 nm for TKB-Ketazine-COF and by 45 nm for TKBPB-Ketazine-COF compared to their corresponding monomers again indicate that the ketazine linkage enables π conjugation. From the absorption spectra, the optical bandgap was evaluated to be 2.74 and 2.64 eV for TKB-Ketazine-COF and TKBPB-Ketazine-COF, respectively, by the Kubelka–Munk method (Figures S43 and S44, Table S18). Cyclic voltammetry revealed that TKB-Ketazine-COF (Figure S45) and TKBPB-Ketazine-COF (Figure S46) have oxidation potentials of 0.43 and 0.44 V and reduction

potentials of -1.20 and -1.10 V, respectively. Therefore, the electrochemical bandgap is 1.63 and 1.54 eV for TKB-Ketazine-COF and TKBPB-Ketazine-COF, respectively. The highest occupied molecular orbital (HOMO) and the lowest unoccupied molecular orbital (LUMO) are -5.02 and -3.39 eV and -5.03 and -3.49 eV for TKB-Ketazine-COF and TKBPB-Ketazine-COF, respectively (Table S18 and Figure S47).

The solid sample of TKB-Ketazine-COF emits blue centered at 481 nm (Figure 4K, solid blue curve) with an absolute quantum yield of $0.3 \pm 0.1\%$ upon excitation at 364 nm. Similarly, TKBPB-Ketazine-COF emits sky blue light at 483 nm (Figure 4L, solid red curve) with an absolute quantum yield of $0.5 \pm 0.1\%$ upon excitation at 394 nm. The fluorescence lifetime of TKB-Ketazine-COF and TKBPB-Ketazine-COF in the solid state was estimated to be 1.28 and 1.69 ns, respectively (Figures S48 and 49, Table S16). These investigations revealed that the ketazine linkage enables π conjugation between knot and linker and develops highly blue luminescent materials with remarkable fluorescence quantum yields which are key to light-emitting materials and sensing systems. Notably, blue emissive COFs were rarely developed so far.^{22–24}

CONCLUSION

In summary, we have reported the first examples of ketazine-linked COFs. By exploring C_3 -symmetric ketone monomers TKB and TKBPB with different molecular lengths, we have successfully synthesized microporous and mesoporous ketazine-linked COFs in high yields under solvothermal conditions. Notably, the ketazine-linked COFs are thermally stable up to 350 °C under nitrogen and chemically stable under different conditions.

In detail, we elaborated the ketazine linkage into six different aspects: (1) In terms of structural development, the ketazine linkage is versatile for the synthesis of both microporous and mesoporous COFs. (2) With regard to π electronic structural evolution, the ketazine linkage offers π -conjugated structures where the knot and linker units can be electronically coupled, enabling progressed extension of π conjugation over the x - y plane. (3) With regard to emission colors, the ketazine linkage enables the red-shift of the emission band to emit deep or sky-blue light. (4) For light-emitting functions, the ketazine linkage endows the resultant COFs with impressive quantum yields. (5) With regard to fluorescence lifetime, the ketazine linkage enables COFs to keep photoexcited state over an exceptionally long period. The disclosure of these fundamental aspects of ketazine linkage offers the basis for the further exploration of chemistry and materials of ketazine-linked COFs.

Our new findings of using ketones as building blocks to construct COFs bring new linkage chemistry and novel structures to the field. With the broad availability of ketone monomers with different π backbones and symmetries, we envision that the ketazine linkage chemistry opens an avenue to the designed synthesis of various COFs with unprecedented structures and properties.

ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/jacs.4c08231>.

Methods and characterizations, materials and synthetic procedures, including experimental setups, characterization of the COFs, the stability of the COFs, and additional data about the photochemical properties of the COFs; Figures S1–S49, Tables S1–S18, and References S1–S12 (PDF)

Atomic coordinates (ZIP)

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Author Contributions

The manuscript was written through contributions of all authors. All authors have given approval to the final version of the manuscript.

Notes

The authors declare no competing financial interest.

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REFERENCES

- (1) Tan, K. T.; Ghosh, S.; Wang, Z.; Wen, F.; Rodríguez-San-Miguel, D.; Feng, J.; Huang, N.; Wang, W.; Zamora, F.; Feng, X.; Thomas, A.; Jiang, D. Covalent Organic Frameworks. *Nat. Rev. Methods Primers* **2023**, 3, 1.
- (2) Diercks, C. S.; Yaghi, O. M. The Atom, The Molecule, and The Covalent Organic Framework. *Science* **2017**, 355, No. eaal1585.
- (3) Liu, R.; Tan, K. T.; Gong, Y.; Chen, Y.; Li, Z.; Xie, S.; He, T.; Lu, Z.; Yang, H.; Jiang, D. Covalent Organic Frameworks: An Ideal Platform for Designing Ordered Materials and Advanced Applications. *Chem. Soc. Rev.* **2021**, 50, 120–242.

- (4) Geng, K.; He, T.; Liu, R.; Dalapati, S.; Tan, K. T.; Li, Z.; Tao, S.; Gong, Y.; Jiang, Q.; Jiang, D. Covalent Organic Frameworks: Design, Synthesis, and Functions. *Chem. Rev.* **2020**, *120*, 8814–8933.
- (5) Jiang, D. Covalent Organic Frameworks: An Amazing Chemistry Platform for Designing Polymers. *Chem.* **2020**, *6*, 2461–2483.
- (6) Dalapati, S.; Jin, S.; Gao, J.; Xu, Y.; Nagai, A.; Jiang, D. An Azine-Linked Covalent Organic Framework. *J. Am. Chem. Soc.* **2013**, *135*, 17310–17313.
- (7) Li, Z.; Feng, X.; Zou, Y.; Zhang, Y.; Xia, H.; Liu, X.; Mu, Y. A 2D Azine-Linked Covalent Organic Framework for Gas Storage Applications. *Chem. Commun.* **2014**, *50*, 13825–13828.
- (8) (a) Vyas, V. S.; Haase, F.; Stegbauer, L.; Savasci, G.; Podjaski, F.; Ochsenfeld, C.; Lotsch, B. V. A Tunable Azine Covalent Organic Framework Platform for Visible Light-Induced Hydrogen Generation. *Nat. Commun.* **2015**, *6*, 8508. (b) Guan, P.; Qiu, J.; Zhao, Y.; Wang, H.; Li, Z.; Shi, Y.; Wang, J. A Novel Crystalline Azine-Linked Three-Dimensional Covalent Organic Framework for CO₂ Capture and Conversion. *Chem. Commun.* **2019**, *55*, 12459–12462.
- (9) Li, Z.; Zhi, Y.; Feng, X.; Ding, X.; Zou, Y.; Liu, X.; Mu, Y. An Azine-Linked Covalent Organic Framework: Synthesis, Characterization and Efficient Gas Storage. *Chem.—Eur. J.* **2015**, *21*, 12079–12084.
- (10) Alahakoon, S. B.; Thompson, C. M.; Nguyen, A. X.; Occhialini, G.; McCandless, G. T.; Smaldone, R. A. An Azine-Linked Hexaphenylbenzene Based Covalent Organic Framework. *Chem. Commun.* **2016**, *52*, 2843–2845.
- (11) Xu, H.; Gao, J.; Jiang, D. Stable, Crystalline, Porous, Covalent Organic Frameworks as A Platform for Chiral Organocatalysts. *Nat. Chem.* **2015**, *7*, 905–912.
- (12) Li, R. L.; Flanders, N. C.; Evans, A. M.; Ji, W.; Castano, I.; Chen, L. X.; Gianneschi, N. C.; Dichtel, W. R. Controlled Growth of Imine-Linked Two-Dimensional Covalent Organic Framework Nanoparticles. *Chem. Sci.* **2019**, *10*, 3796–3801.
- (13) Feriante, C. H.; Jhulki, S.; Evans, A. M.; Dasari, R. R.; Slicker, K.; Dichtel, W. R.; Marder, S. R. Rapid Synthesis of High Surface Area Imine-Linked 2D Covalent Organic Frameworks by Avoiding Pore Collapse During Isolation. *Adv. Mater.* **2020**, *32*, 1905776.
- (14) Li, Z.; Huang, N.; Lee, K. H.; Feng, Y.; Tao, S.; Jiang, Q.; Nagao, Y.; Irle, S.; Jiang, D. Light-Emitting Covalent Organic Frameworks: Fluorescence Improving via Pinpoint Surgery and Selective Switch-On Sensing of Anions. *J. Am. Chem. Soc.* **2018**, *140*, 12374–12377.
- (15) Uribe-Romo, F. J.; Doonan, C. J.; Furukawa, H.; Oisaki, K.; Yaghi, O. M. Crystalline Covalent Organic Frameworks with Hydrazone Linkages. *J. Am. Chem. Soc.* **2011**, *133*, 11478–11481.
- (16) Zhuang, H.; Guo, C.; Huang, J.; Wang, L.; Zheng, Z.; Wang, H.-N.; Chen, Y.; Lan, Y.-Q. Hydrazone-Linked Covalent Organic Frameworks. *Angew. Chem., Int. Ed.* **2024**, *63*, No. e202404941.
- (17) Nagai, A.; Chen, X.; Feng, X.; Ding, X.; Guo, Z.; Jiang, D. A Squaraine-Linked Mesoporous Covalent Organic Framework. *Angew. Chem., Int. Ed.* **2013**, *125*, 3858–3862.
- (18) Xue, R.; Gou, H.; Zheng, Y.; Zhang, L.; Liu, Y.; Rao, H.; Zhao, G. A New Squaraine-Linked Triazinyl-Based Covalent Organic Frameworks: Preparation, Characterization and Application for Sensitive and Selective Determination of Fe³⁺ Cations. *Chem. Select* **2020**, *5*, 10632–10636.
- (19) Jin, E.; Asada, M.; Xu, Q.; Dalapati, S.; Addicoat, M. A.; Brady, M. A.; Xu, H.; Nakamura, T.; Heine, T.; Chen, Q.; Jiang, D. Two-Dimensional sp² Carbon-Conjugated Covalent Organic Frameworks. *Science* **2017**, *357*, 673–676.
- (20) Han, X.; Zhou, Z.; Wang, K.; Zheng, Z.; Neumann, S. E.; Zhang, H.; Ma, T.; Yaghi, O. M. Crystalline Polyphenylene Covalent Organic Frameworks. *J. Am. Chem. Soc.* **2024**, *146*, 89–94.
- (21) Wang, Y.; Hao, W.; Liu, H.; Chen, R.; Pan, Q.; Li, Z.; Zhao, Y. Facile Construction of Fully sp²-Carbon Conjugated Two-Dimensional Covalent Organic Frameworks Containing Benzobisthiazole Units. *Nat. Commun.* **2022**, *13*, 100.
- (22) Gao, Q.; Li, X.; Ning, G.; Leng, K.; Tian, B.; Liu, C.; Tang, W.; Xu, H.; Loh, K. Highly Photoluminescent Two-Dimensional Imine-

Based Covalent Organic Frameworks for Chemical Sensing. *Chem. Commun.* **2018**, *54*, 2349–2352.

(23) Wan, S.; Guo, J.; Kim, J.; Ihee, H.; Jiang, D. A Belt-Shaped, Blue Luminescent, and Semiconducting Covalent Organic Framework. *Angew. Chem., Int. Ed.* **2008**, *47*, 8826–8830.

(24) Wan, S.; Guo, J.; Kim, J.; Ihee, H.; Jiang, D. A Photoconductive Covalent Organic Framework: Self-Condensed Arene Cubes Composed of Eclipsed 2D Polypyrene Sheets for Photocurrent Generation. *Angew. Chem., Int. Ed.* **2009**, *48*, 5439–5442.



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